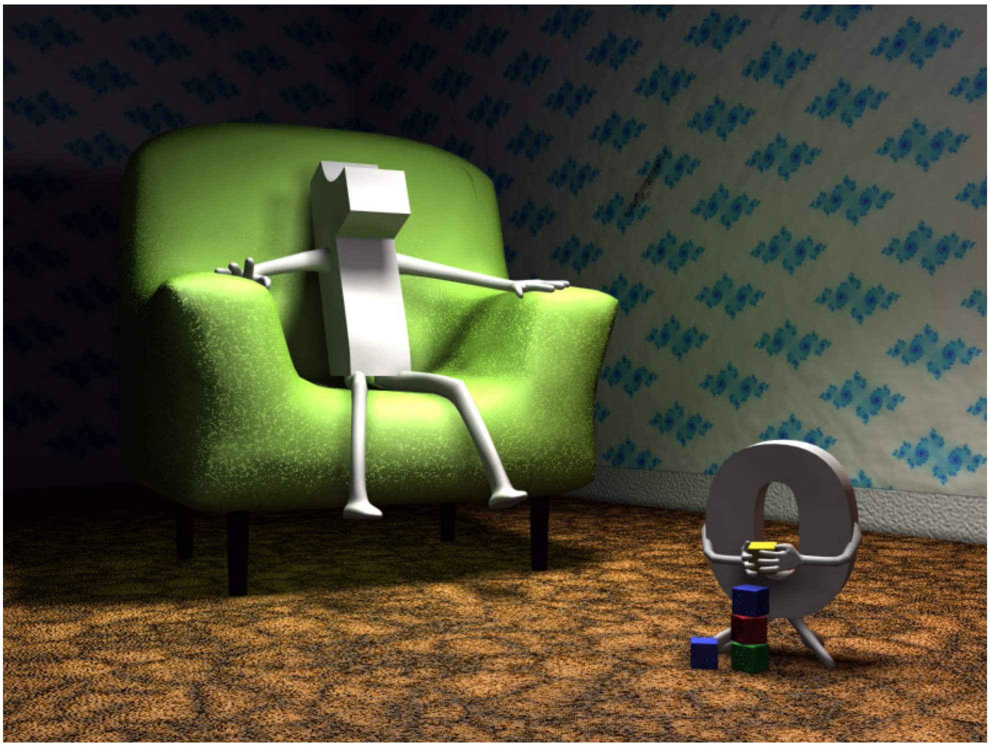
Learning how to prove Workshop Week 2

Integers



Prime Factorisation. Integer modulo n

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1. **Let a and b be positive integers. If $a = a_1 c$ and $b = b_1 c$, where $c = \gcd(a, b)$, then $\operatorname{lcm}(a, b) = a_1 b_1 c$.**

> **Solution:** We know that $ab = \gcd(a, b)\operatorname{lcm}(a, b)$. Keeping in mind that $a = a_1 c$, $b = b_1 c$ and $c = \gcd(a, b)$ we get that
>
> $$(a_1 c)(b_1 c) = c\,\operatorname{lcm}(a, b),$$
>
> which implies that $\operatorname{lcm}(a, b) = a_1 b_1 c$.

2. **Find all solutions in $\mathbb{Z}_n$ (as indicated) for each of the given equations:**

$$\text{(a)} \quad \left.\begin{aligned} \bar{3}x + \bar{2}y &= \bar{1} \\ \bar{5}x + y &= \bar{1} \end{aligned}\right\} \text{ in } \mathbb{Z}_{11}, \qquad \text{(b)} \quad \left.\begin{aligned} \bar{3}x + \bar{2}y &= \bar{1} \\ \bar{5}x + y &= \bar{4} \end{aligned}\right\} \text{ in } \mathbb{Z}_{7}$$

> **Solution:** (a)
>
> $$\left.\begin{aligned} \bar{3}x + \bar{2}y &= \bar{1} \\ \bar{5}x + y &= \bar{1} \end{aligned}\right\} \text{ in } \mathbb{Z}_{11}$$
>
> Multiply both sides of the second equation by $\bar{2}$ to get $\overline{10}x + \bar{2}y = \bar{2}$, and subtract the first equation from it: $\bar{7}x = \bar{1}$.
>
> Next, we compute the inverse of $\bar{7}$ in $\mathbb{Z}_{11}$. Notice that such an inverse exists since 7 and 11 are coprime. Proceeding like in the previous exercise and apply the Euclidean Algorithm to get
>
> $$\begin{aligned} 11 &= 1 \cdot 7 + 4, \\ 7 &= 1 \cdot 4 + 3, \\ 4 &= 1 \cdot 3 + 1, \end{aligned}$$
>
> which implies that
>
> $$\begin{aligned} 1 &= 4 - 1 \cdot 3 = 4 - 1 \cdot (7 - 1 \cdot 4) = (-1)7 + 2 \cdot 4 = (-1)7 + 2(11 - 1 \cdot 7) = \\ &= (-3) \cdot 7 + 2 \cdot 11. \end{aligned}$$
>
> Thus $\bar{7}^{-1} = \overline{-3} = \bar{8}$. From here we get that
>
> $$\bar{7}x = \bar{1} \Rightarrow x = \bar{8} \text{ in } \mathbb{Z}_{11}.$$

Therefore: $y = \bar{1} - \bar{5}x = \bar{1} - \bar{5} \cdot \bar{8} = \bar{1} - \overline{40} = \overline{-39} = \overline{-6} = \bar{5}$

(b)

$$\left.\begin{aligned} \bar{3}x + \bar{2}y &= \bar{1} \\ \bar{5}x + y &= \bar{4} \end{aligned}\right\} \text{ in } \mathbb{Z}_7$$

Multiply both sides of the second equation by $\bar{2}$:

$$\overline{10}x + \bar{2}y = \bar{8} \Rightarrow \bar{3}x + \bar{2}y = \bar{1},$$

which coincides with the first equation. Thus, we only need to solve the second equation, namely:

$$\bar{5}x + y = \bar{4}.$$

If $x = \bar{r}$ is an element of $\mathbb{Z}_7 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}\}$, then

$$y = \bar{4} - \bar{5}x = \bar{4} - \bar{5} \cdot \bar{r} = \bar{4} - \overline{5r}.$$

Thus the solutions are the following:

x	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{6}$
y	$\bar{4}$	$\bar{6}$	$\bar{1}$	$\bar{3}$	$\bar{5}$	$\bar{0}$	$\bar{2}$

For example, in $\mathbb{Z}_7$ for $x = \bar{0}$, we get that $y = \bar{4}$; for $x = \bar{1}$, we have that $y = \bar{4} - \bar{5} = \overline{-1} = \bar{6}$, and so on.